26020 Chemistry - Polytechnic Foundation winter 2024/Kemi Polyteknisk Grundlag Vinter 2024 - ikke randomiseret

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)
"." bruges som decimalseparator på både dansk og engelsk.
Der er i beregnede svar brugt heltallige molvægte af grundstofferne hydrogen (1 g/mol), kulstof (12g/mol) og ilt (16/mol).

The following approach to scoring responses is implemented and is based on "One best answer"

There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point

Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)
”.” is used as decimal separator in the both Danish and English text.
In calculated answers integer molecular weights are used for the elements, hydrogen (1 g/mol), carbon (12 g/mol), and oxygen (16 g/mol).

I hvilken af nedenstående opstillinger af grundstofferne Na, Al, P og Cl fra det periodiske systems 3. periode er atomradius faldende gennem rækken?

In which of the following arrangements of elements Na, Al, P, and Cl from the third period are the atomic radius decreasing through the series?

Choose one answer

- Na, P, Al, Cl
- Al, Na, Cl, P
- Cl, P, Al, Na
- Na, Al, P, Cl
- P, Al, Na, Cl

Hvilket symbol for isotopen af klor med atomvægt 37 u er korrekt?

Which representation of the isotope of chlorine with atomic mass 37 u is correct ?

Choose one answer

- Cl_{17}^{37}
- ${}^{37}_{17}\mathrm{Cl}$
- ${}^{17}_{37}\mathrm{Cl}$
- Cl_{37}^{17}
- ${}^{37}\mathrm{Cl}_{17}$

Fruktose fermenteres efter følgende ikke balancerede reaktionsskema:
$C_6H_{12}O_6 \rightarrow CO_2 + CH_3CH_2OH$
Hvor mange mol og gram ethanol dannes når der startes med 1 kg fruktose?

Fructose is fermented according to the following unbalanced reaction scheme:
$C_6H_{12}O_6 \rightarrow CO_2 + CH_3CH_2OH$
How many moles and grams of ethanol is produced when starting from 1 kg of fructose?

Choose one answer

- 0 mol, 0 g
- 10.9 mol, 500 g
- 5.6 mol, 255.6 g
- 21.7 mol, 1000 g
- 11.1 mol, 511 g

I en 10 l tank reageres 10 g $\mathrm{H_2}$ og 16 g $\mathrm{O_2}$ ved 25 $^o\mathrm{C}$ og temperaturen stiger til 1950 $^o\mathrm{C}$. Hvad er trykket i tanken?

In a 10 l tank 10g $\mathrm{H_2}$ and 16 g $\mathrm{O_2}$ at 25 $^o\mathrm{C}$ is reacted and forms water. Assume that the temperature rises to 1950 $^o\mathrm{C}$. What is the pressure in the tank?

Choose one answer

- 9.24 MPa
- 10.2 MPa
- 12.0 MPa
- 91.2 MPa
- 8.11 MPa

For reaktionen

$2\mathrm{Al}(s) + 3\mathrm{Br}_2(l) \rightarrow 2\mathrm{AlBr}_3(s)$

er

$\Delta G^0 = -1080.2\ \mathrm{kJmol}^{-1}$ og $\Delta S^0 = -180\ \mathrm{Jmol}^{-1}\mathrm{K}^{-1}$.

Hvad er ΔH^0 for reaktionen ved 50 $^o\mathrm{C}$?

For the reaction

$2\mathrm{Al}(s) + 3\mathrm{Br}_2(l) \rightarrow 2\mathrm{AlBr}_3(s)$

$\Delta G^0 = -1080.2\ \mathrm{kJmol}^{-1}$ and $\Delta S^0 = -180\ \mathrm{Jmol}^{-1}\mathrm{K}^{-1}$.

What is ΔH^0 for the reaction at 50 $^o\mathrm{C}$?

Choose one answer

- -1022 kJ/mol
- -1071 kJ/mol
- -1138 kJ/mol
- 57087 kJ/mol
- Det kan ikke beregnes
  you cannot calculate

For en reaktion gælder at både ΔH og ΔS er positive. Hvilket af følgende udsagn om reaktionen er korrekt?

For a reaction both ΔH og ΔS are positive. Which statement about the reaction is correct?

Choose one answer

- Reaktionen forløber spontant ved alle temperature.
  The reaction is spontaneous at all temperatures.
- Reaktionen forløber spontant ved lav temperatur men ikke spontant ved høj temperatur.
  The reaction is spontaneous at low temperature but not spontaneous at high temperature.
- Reaktionen forløber aldrig spontant.
  The reaction is never spontaneous.
- Reaktionen forløber ikke spontant ved lav temperatur men spontant ved høj temperatur.
  The reaction is not spontaneous at low temperature but spontaneous at high temperature.
- Det er ikke muligt at udtale sig om ved hvilke temperaturer reaktionen forløber spontant.
  It is not possible to make statements about at which temperatures the reaction is spontaneous

Bortset fra spin hvilke af følgende sæt af kvantetal karakteriserer en 4d elektron i iod.

Disregarding spin, which of the following set of quantum numbers are valid for a 4d electron in iodine?

Choose one answer

- $n = 4$; $l = 2$; $m = 0$
- $n = 4$; $l = 2$; $m = 4$
- $n = 1$; $l = 2$; $m = 1$
- $n = 3$; $l = 4$; $m = 0$
- $n = 5$; $l = 2$; $m = 0$

Ædelgasserne He, Ne, Ar, Kr, Xe har følgende respektive antal elektroner i den yderste skal:

The nobel gasses He, Ne, Ar, Kr, Xe have the following number of electrons in the outer shell, respectively:

Choose one answer

- 2, 8, 8, 8, 8
- 8, 8, 8, 8, 8
- 8, 8, 8, 8, 2
- 2, 8, 18, 32, 50
- 2, 2, 2, 2, 2

Hvilke af følgende molekyler har en plan geometri: NH_3, SF_6, BH_3 og CH_4?

Which of the following molecules have planar geometry: NH_3, SF_6, BH_3, and CH_4?

Choose one answer

- både BH_3 og NH_3
  both BH_3 and NH_3
- kun BH_3
  only BH_3
- ingen af dem
  none of them
- dem alle sammen
  all of them
- kun NH_3
  only NH_3

I argon i fast form er bindingerne mellem argon atomerne:

In solid argon, the bonding between the argon atoms is:

Choose one answer

- kovalente bindinger
  covalent bonds
- intermolekylære bindinger
  intermolecular bonds
- ioniske bindinger
  ionic bonds
- metalbindinger
  metal bonds
- hydrogenbindinger
  hydrogen bonds

12 g natriumbromid opløses i 300 ml vand. Hvad er frysepunktet af denne opløsning
Brug 1 kg/l som vands densitet, 0 $^o\mathrm{C}$ som rent vands frysepunkt og
$K_f = 1.86^o\mathrm{C/m}.$

12 g of sodium bromide is dissolved in 300 ml of water. What is the freezing point of this solution.
Use 1 kg/l for the density of water, 0 $^o\mathrm{C}$ as the freezing point of pure water and
$K_f = 1.86^o\mathrm{C/m}.$

Choose one answer

- 1.45$^o\mathrm{C}$
- -1.45$^o\mathrm{C}$
- -0.72$^o\mathrm{C}$
- 0.72$^o\mathrm{C}$
- 0$^o\mathrm{C}$

Rent jern krystalliserer i et rumcentreret kubisk gitter. Givet at densiteten af jern er 7874 $\mathrm{kg/m^3}$ ved stuetemperatur og atomvægten 55.85 g/mol hvad er volumen af enhedscellen i krystallen?

Pure iron crystallizes in a body centered cubic lattice. Given that the density is 7874 $\mathrm{kg/m^3}$ at room temperature and the atomic mass 55.85 g/mol, what is the volume of the unit cell in the crystal?

Choose one answer

- $2.36 \times 10^{-29}\ \mathrm{m^3}$
- $1.18 \times 10^{-29}\ \mathrm{m^3}$
- $2.36 \times 10^{-26}\ \mathrm{m^3}$
- $4.71 \times 10^{-29}\ \mathrm{m^3}$
- $1.18 \times 10^{-26}\ \mathrm{m^3}$

I et eksperiment hvor iodid oxideres i sur opløsning af hydrogenperoxid følges koncentration af iodid som funktion af tiden. De nedenfor given data registreres. Hvad er reaktionsordenen i koncentrationen af iodid?

In an experiment where iodide is oxidized in acidic solution the concentration of iodide is measured as a function of time. The data given below is obtained. What is the reaction order in iodide concentration?

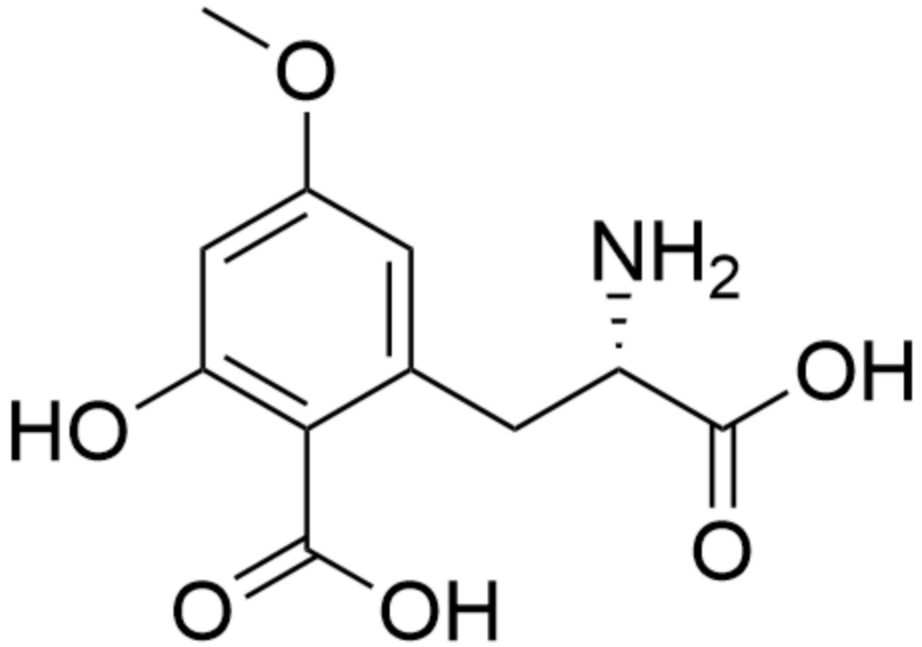
t/s	$[I^-]$/M
0	1
60	0.498
120	0.250
300	0.0312
1000	9.59×10^{-6}

Choose one answer

- ½
- 2
- 1.5
- Det er ikke muligt at bestemme
  You cannot tell
- 1

Stoffet vist nedenfor findes i stjernefrugt som gror naturligt i Sydøstasien. Hvilke af følgende liste af funktionelle grupper indeholder stoffet og er det chiralt/ ikke chiralt:

The compound shown below is found in starfruit which is naturally occuring in Southeast Asia. Which of the following list of functional groups is found in the compound and is it chiral/non chiral:



Choose one answer

- Ester, hydroxyl, carboxyl, amin. Stoffet er ikke chiralt.
  Ester, hydroxy, carboxyl, amine. The compound is not chiral.
- Ether, hydroxyl, carboxyl, amin. Stoffet er chiralt.
  Ether, hydroxy, carboxyl, amine. The compound is chiral.
- Ether, keton, hydroxyl, amin. Stoffet er ikke chiralt.
  Ether, ketone, hydroxy, amine. The compound is not chiral.
- Ether, hydroxyl, carboxyl, amin. Stoffet er ikke chiralt.
  Ether, hydroxy, carboxyl, amine. The compound is not chiral.
- Ether, keton, hydroxyl, amin, Stoffet er chiralt.
  Ether, ketone, hydroxy, amine. The compound is chiral.

Nitrogen og hydrogen reageres ved hjælp af en katalysator ved 455 $^{o}\mathrm{C}$ og danner ammoniak. Den ikke balancerede reaktionsligning er
$\mathrm{N_2 + H_2 \rightleftharpoons NH_3}$.
I en tryktank tilsættes kvælstof og hydrogen. Før reaktionen startes er partialtrykkene af nitrogen og hydrogen
$p_{N_2} = 100$ bar og $p_{H_2} = 300$ bar.
Når reaktion er nået til ligevægt er $p_{NH_3} = 95$ bar.
Hvad er ligevægtskonstanten, K_p?

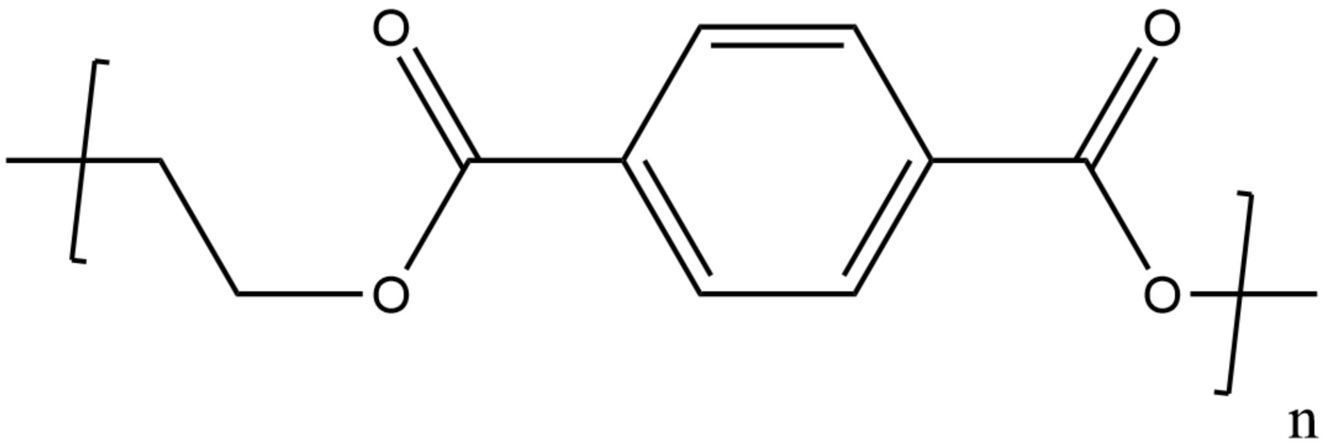
Nitrogen and hydrogen are reacted using a catalyst at 455 $^{o}\mathrm{C}$ and forms ammonia. The unbalanced reaction equation is
$\mathrm{N_2 + H_2 \rightleftharpoons NH_3}$.
In a pressure vessel nitrogen and hydrogen are introduced. Before the reaction starts the partial pressures of nitrogen and hydrogen are
$p_{N_2} = 100$ bar and $p_{H_2} = 300$ bar.
When the reaction has reached equilibrium, $p_{NH_3} = 95$ bar.
What is the equilibrium constant, K_p?

Choose one answer

- 2.27×10^{4}
- 1.15×10^{-2}
- 4.40×10^{-5}
- 8.70×10^{1}
- 1.30×10^{-7}

En prøve af polymeren PET (polyethylenterephthalat) som vist nedefor har molekylvægten 100 kg/mol. Hvor stor er polymerisationsgraden (*n* i nedenstående formel)?

A sample of the polymer PET (polyethyleneterephthalate) as shown below has the molecular weight 100 kg/mol. What is the degree of polymerization (*n* in the formula below)?



Choose one answer

- 1042
- 521
- 515
- 1030
- 0.5

Der fremstilles en opløsning af 19.6 g 100% fosforsyre og 10 g natriumhydroxid i 200 ml vand.
Hvad er pH i denne opløsning?
De 3 syrestyrkekonstanter for fosforsyre er
$K_{a,1} = 7.5 \times 10^{-3}; K_{a,2} = 6.2 \times 10^{-8}; K_{a,3} = 4.8 \times 10^{-13}$.

A solution is made by dissolving 19.6 g of 100% phosphoric acid and 10 g of sodium hydroxide in 200 ml of water.
What is pH in this solution?
The three ionization constants for phosphoric acid are
$K_{a,1} = 7.5 \times 10^{-3}; K_{a,2} = 6.2 \times 10^{-8}; K_{a,3} = 4.8 \times 10^{-13}$.

Choose one answer

- 7.68
- 7.21
- 2.60
- 12.80
- 6.73

Der fremstilles en mættet opløsning af magnesiumhydroxid ($Mg(OH)_2$) i vand.
Magnesiumhydroxid er tungt opløseligt –opløselighedsproduktet for $Mg(OH)_2$ er $K_{sp} = 5.61 \times 10^{-12}$ ved 25 $^o\mathrm{C}$.
Hvad er pH i den mættede magnesiumhydroxid opløsning?

A saturated solution of magnesium hydroxide ($Mg(OH)_2$) in water is prepared.
Magnesiumhydroxide is only slightly soluble in water – the solubility product for $Mg(OH)_2$ is $K_{sp} = 5.61 \times 10^{-12}$ at 25 $^o\mathrm{C}$.
What is pH in the saturated magnesium hydroxide solution?

Choose one answer

- 10.25
- 10.55
- 8.27
- 10.35
- 11.25

En elektrokemisk celle indeholder to identiske nikkelelektroder i to nikkel(II)-opløsninger som er forbundet med en saltbro. Ni^{2+} koncentrationerne er henholdsvis 0.5 M og 3×10^{-5} M i de 2 opløsninger.
Hvad er den elektromotoriske kraft for den spontant forløbende proces i cellen, E_{cell}?

An electrochemical cell contains two identical nickel electrodes in two nickel solutions which are connected by a salt bridge. The Ni^{2+} concentration in the two solution are 0.5 M and 3×10^{-5} M , respectively.
What is the electromotive force of the spontaneous process in the cell, E_{cell}?

Choose one answer

- -125mV
- -250mV
- 125mV
- 143mV
- 250mV

Når den følgende reaktion er afstemt med det numerisk mindst mulige set af heltal, hvad er summen af koefficienterne (glem ikke koefficienter af 1):

$$MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2 \text{(i sur opløsning)}$$

When the following reaction is balanced with the smallest possible set of integers, what is the sum of all the coefficients (do not forget coefficients of 1):

$$MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2 \text{(in acidic solution)}$$

Choose one answer

- 19
- 43
- 40
- 7
- 35